ECON 573 Machine Learning in Econometrics
Group Project

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1 Introduction

The U.S Banking sector has observed substantial changes in depositor behavior during periods of increases in interest rates in the past. In March 2022, in response to the COVID-19 pandemic and excessive inflationary pressures, the Federal Reserve tightened monetary policies by increasing the target range of the federal funds rate (FFR) from 0.00% to 0.25 to 4.75% to 5.00%. With the gradual transmission of the federal funds rate to market interest rates of various maturities, depositors sought out alternative investment opportunities with more attractive returns. This precipitated a substantial departure of deposits from the banking system. During the second and third quarters of 2022, the United States banking industry's deposits decreased by $573 billion, according to data from the S&P Global Market Intelligence Database. This trend has continued through the fourth quarter of 2022 and into 2023, as evidenced by the Federal Reserve's weekly data.

Historically, the United States has encountered numerous bank liquidity crises related to deposit outflows. For instance, during the Savings and Loan Crisis in the 1980s, a combination of factors, including rising interest rates, deregulation, and the outflow of deposits, have led to the collapse of a number of savings and loan associations. Similarly, the 1982 failure of Penn Square Bank was partially attributable to deposit withdrawals caused by excessive interest rates. Given the importance of banks to the economy and the significant effect that deposit withdrawals have on their stability, it is vital to gain a thorough knowledge of this phenomenon under high-interest-rate environments.

In this project, we utilized a variety of machine learning techniques to estimate the impact of interest rates on a bank's total deposits. We compiled multidimensional panel data of over one million banks' deposit levels that were measured quarterly from December 1992 to June 2022 and obtained from the Federal Financial Institutions Examination Council's (FFIEC) call report data. We employed Multiple Linear Regression, Polynomial Regression, Ridge Regression, The Lasso, Natural Cubic Spline, Generalized Additive Models, and Principal Component Regression. These models consistently demonstrated a negative relationship between interest rates and deposit levels, implying that deposit outflows occur in environments with high interest rates. To optimize model parameters, we utilized cross-validation techniques. In addition, we examined the trade-off between model interpretability and prediction accuracy in order to select the model that best depicts the relationship between interest rates and bank deposit levels. Our study sought to provide valuable insights into the dynamics of deposit outflows in response to changes in interest rates, informing strategies to strengthen the resilience of the U.S. banking sector.

2 Literature review

An empirical study by Drechsler, Savov, and Schnabl (2017) proposed a deposit channel of monetary policy to examine the effect of change in federal funds rate on the aggregate deposit level. They hypothesized that agents in the economy hold three assets: cash offering zero interest rates, saving deposits offering low-interest rates, and bonds offering interest rate equivalent to the federal funds rate. Bond yields tend to adjust simultaneously with the policy rate, while deposit rates barely change and remain relatively low even in a high-interest rate environment. Therefore, when the federal funds rate increases, the spread between bond yields and deposit rate widens, which signifies higher opportunity costs of holding deposits or cash. Such deposits outflow from the banking system to the financial market is described as the deposit channel of monetary policy.

Moreover, economist Rafael Repullo (2020) critiqued the model proposed by Drechsler et al., proposing that interest rates have ambiguous effects on deposits level. He demonstrated a U-shaped relationship between deposits and interest rates, where deposits first decrease but then increase as interest rates rise. In his model, the effect of an increase in the policy rate on equilibrium deposits can be decomposed into a negative substitution effect due to an increase in the opportunity cost of holding deposits and a positive income effect due to a higher return to the household's financial wealth. When the policy rate is near zero, cash and deposits yield the same return as bonds. Due to the liquidity services banks provide, households tend to invest most of their wealth in cash or deposits and decrease that investment when the policy rate rises due to the negative substitution effect. As the interest rate reaches a certain high level, the income effect increases the aggregate deposit level.

Deposits compose around 75% of bank balance sheets (Gerlach et al., 2017). The stability of deposit inflow allows banks to expand illiquid assets, particularly small business loans, so decreases in deposits induce the reduction in lending services and exacerbate the tightening effect of monetary policies (Gerlach et al., 2017). To recover the loss in demand deposits, banks have to offer higher payment on interest-bearing deposits, which also increases banks funding costs. High-interest rates also cause depreciation in the present value of banks' assets along with a deposit outflow, which further intensifies bank insolvency risks. The deposit channel of monetary policy could have a profound impact and contribute to the systematic instability of the banking industry.

3 Connecting Research Question with An Empirical Model

For bank i and at time t, we develop the following econometric model to estimate the effect of interest rates on its deposit level:

$$\begin{aligned}\log(\text{Total Deposit}_{it}) = \beta_0 + \beta_1 \text{FFR}_t + \beta_2 \text{10y nom}_t + \beta_3 \text{year}_{it} + \sum_{j=1}^{p} \gamma_j Z_{ji} \\ + \beta_4 \text{Umich Consumer Sent}_t + \beta_5 \text{Real Disposable Income}_t + \beta_6 \text{Population}_t \\ + \beta_7 \text{Average Bank Capital Size}_t + \beta_8 \text{NASDAQ}_t + \epsilon_{it},\end{aligned} \tag{1}$$

where Total Deposit$_{it}$ is the total deposit held by bank i at time t; our main predictor is FFR$_t$, the federal funds rate set by the Fed, a proxy for short-term interest rates, while 10y nom$_t$ is the

market yield on U.S. Treasury securities at 10-year maturity, a proxy for long-term interest rates. year_t is a categorical variable with levels 1 to 31 corresponding to year 1992 to 2022. The term $\sum_{j=1}^{p} \gamma_j Z_{ji}$ represents different time-invariant bank-level properties, including the region that bank i is located in, what does bank i specialize in (commercial, agricultural, mortgage, credit card, etc.), total number of domestic offices, whether it is supervised by the Fed, FDIC, or OCC, and other bank characteristics. Umich Consumer Sent_t is the consumer sentiment index constructed by the University of Michigan, included as a control variable because consumer confidence about the economy may alter depositors willingness to spend and precautionary saving demand. Real $\text{Disposable Income}_t$, Population_t, and Average Bank Capital Size_t are also critical factors that affect the banks' deposit base, so they were included as control variables. NASDAQ_t, the NASDAQ Composite Index, a proxy for the performance of the U.S. stock market, was also included since a rising stock market could lead to an increase in investors' financial health and deposit demand.

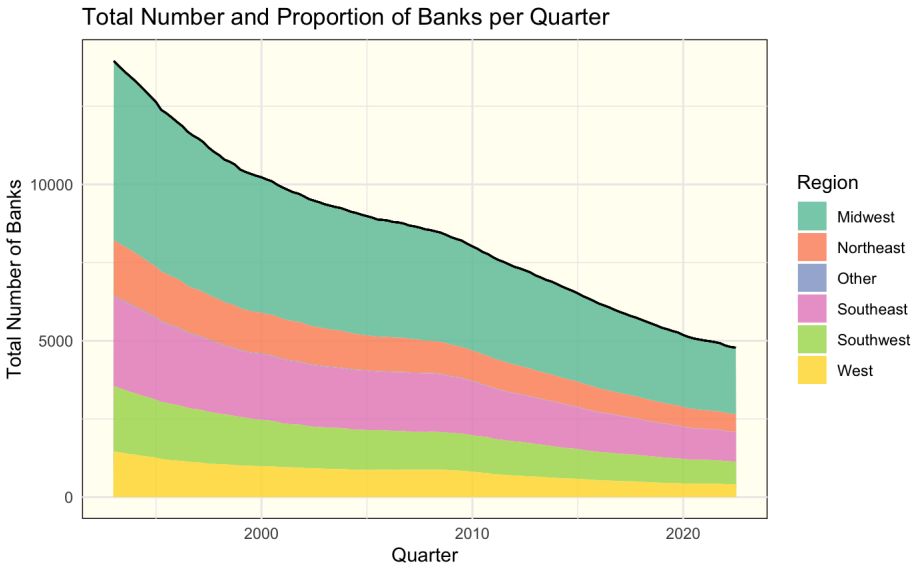
4 Data Description

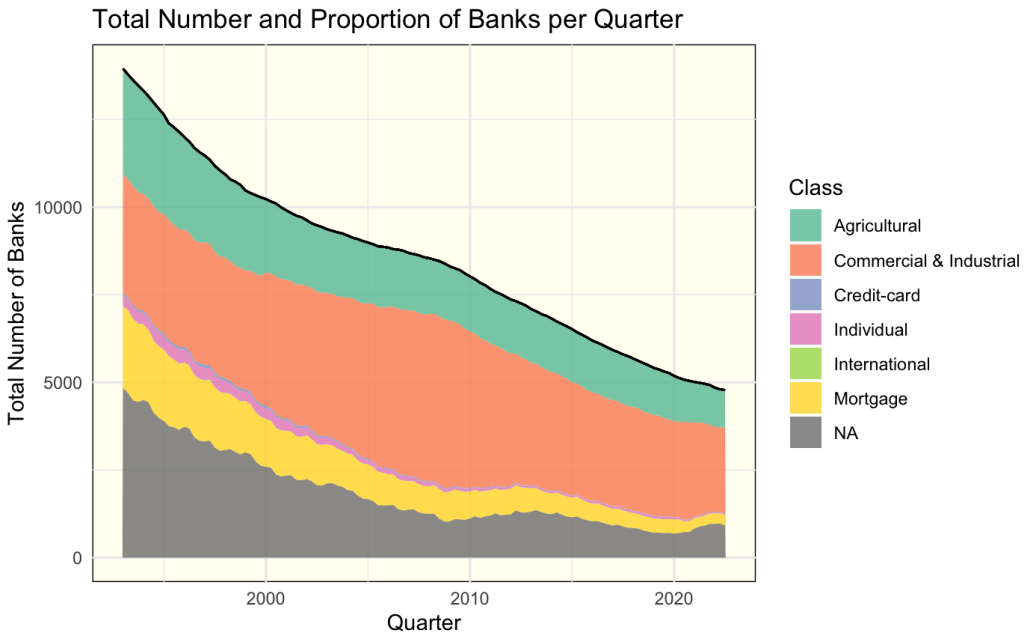
We compiled an extensive panel data set containing information on banks' deposit compositions. Our data set contains 1,014,155 banks' deposit levels collected on a quarterly basis, spanning from December 1992 to June 2022. The source of our data is the FFIEC Call Report, which provides a reliable and comprehensive account of banks' financial activities. Finally, the deposit level was adjusted for inflation and converted to real values. Below is a brief summary statistic table of all the quantitative variables in the dataset.

Table 1: Summary Statistics of Quantitative Variables

	Statistics				
Variable	Mean	Std.Dev	Min	Max	N.Valid
Total Deposits	4710.18	87608.46	0.00	9258432.00	1014155
FFR	2.83	2.20	0.06	6.52	1014155
10-year nom	4.40	1.75	0.65	7.84	1014155
NASDAQ	2982.28	2743.86	625.56	15390.79	1014155
Offdom	10.55	98.90	0.00	6729.00	1014155
Offfor	0.10	4.31	0.00	330.00	1014155
Population	294022.91	22524.65	258276.67	333320.33	1014155
Real Disposable Income	10622.96	2401.23	7225.70	17325.27	1014155
Average Bank Equity Capital	144.53	120.52	24.22	486.37	1014155
Umich Consumer Sent	88.41	12.12	57.67	110.13	1014155

The two figures below are are stacked area plot with the vertical axis represent total number of banks, and horizontal axis are quarters from 1992 to 2022. We could easily observe a decrease in the total number of banks over time, probably because of slower new banks entries, more merges and acquisitions, bank failures, and more stringent regulations.





In the first plot, areas with different colors on the graph represent the proportions of banks belongs to different regions. In the second plot, those areas represent the proportion of banks specializing in commercial, industrial, agricultural, etc. The "Northeast" and "South" regions consistently have the highest proportions of banks, followed by the "Midwest" and "West" regions. The proportion of banks in the "Commercial & Industrial," "Agricultural," "Individual," and "Mortgage" categories remains relatively consistent throughout the years. The "Credit-card" and "International" categories exhibit a small growth in their proportions over time, but still make up a small percentage of the overall number of banks. The "Other" category shows a decrease in proportion over time, which could indicate a shift towards more specialized services.

In general, although the total number of banks decrease over time, The proportions of banks in different regions and specialization groups exhibit some fluctuations, but the overall trends remain relatively stable.

Despite the richness of the data, there are some limitations that need to be acknowledged.

First, our main predictor, the federal funds rate (FFR), has a considerable number of values close to zero. This is a consequence of the Federal Reserve's response to the 2008 financial crisis, during which it lowered the interest rate to the effective lower bound to stimulate the economy. As a result, the FFR may exhibit limited fluctuations after 2008, potentially affecting the robustness of our findings. Nevertheless, we have chosen to use the FFR as our main predictor, since it is generally regarded as an exogenous variable determined solely by the Federal Reserve's decisions, rather than by market conditions. Using alternative interest rates could introduce endogeneity bias into our analysis.

Second, multicollinearity might be an issue in our dataset, as the included variables are macroeconomic indicators that could be highly correlated with each other. Multicollinearity can inflate the variance of the estimated coefficients, making it difficult to identify the individual effects of the predictors on the dependent variable. To reduce multicollinearity, we did not include other short-term interest rate in the model but only FFR and 10-year interest rates as they are less likely to be highly correlated due to the significant difference in maturity.

Lastly, our dataset exhibits a decline in the total number of banks over time. This reduction is not random, as banks have not dropped out of the dataset indiscriminately. This non-random attrition could potentially bias our results and limit the generalizability of our findings.

5 Econometrics analysis and results

In this section, we developed seven models incorporating various machine learning techniques to predict the deposit outflow under high-interest-rate environment. Instead of using non-parametric methods, we opted for parametric approaches as our primary goal was to estimate the effect of interest rates on banks' total deposits, rather than focusing solely on prediction accuracy. The data set under consideration describes the quarterly deposit levels for a varying number of banks, spanning from December 1992 to June 2022.

To create the training and testing data sets using the validation set approach, we employed a random selection of 50% of banks within each quarter for our training data set, while assigning the remaining banks to the testing data set. Following the data cleaning process, the resulting sample sizes for the training and testing data sets were 507,047 and 505,111, respectively. It is important to note that we only utilized the training data set for estimating models and conducting cross-validation in order to choose the optimal model parameters.

5.1 Multiple Linear Regression

The multiple linear regression model was fitted to predict log(Total Deposits) using various predictors. The main predictors of interest are FFR (quarterly federal funds rate) and 10y nom (the

market yield on 10-year Treasury bonds). The model has an adjusted R-squared of 0.456, indicating that the predictors account for about 45.6% of the variability in log(Total Deposits).

The coefficients of FFR and 10y nom are both -0.0110, respectively. This suggests that, holding all other variables constant, a one-unit increase in the short term interest rate or the long term interest rate is associated with a 1.10% decrease in total deposits. Both predictors are statistically significant at the 5% level, with p-values of 0.003 and 0.017, respectively. The estimated coefficient for predictor offdom (number of domestic offices), a proxy for bank size, is also significant. All else held constant, a one-unit increase in offdom increase the bank's total deposit level by 0.3%

Below is a summary table of a few selected predictors:

Variable	**Estimate**	**Std. Error**	**t statistic**	**p-value**
(Intercept)	12.560	0.638	19.693	< 2e-16***
FFR	-0.011	0.004	-2.924	0.003**
10y nom	-0.011	0.005	-2.385	0.017*
offdom	0.003	0.000	204.913	< 2e-16***
Model Statistics				
Residual Std. Error	0.990 on 506991 df			
Multiple R-squared	0.456			
Adjusted R-squared	0.456			
F-statistic	7715 on 55 and 506991 DF			
p-value	< 2.2e-16			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Table 2: Simplified Linear Regression Model Summary for log(Total_Deposits)

When it comes to multicollinearity, there is no direct evidence of it in the provided table. However, multicollinearity can inflate standard errors and impact the stability of the coefficients, decreasing the t-statistic and significance of most of the predictors in the linear model. Thus, we continued exploring other machine learning models as will be discussed moving forward.

Regarding the year predictors, starting from year16 (corresponding to 2008), most of them are statistically insignificant. This suggests that after the financial crisis in 2008, the impact of the time dummies on total deposits is less substantial compared to the years before 2008.

The residuals of the linear model are roughly centered around zero; however, they do not evenly spread on either side with extreme predicted values, suggesting concerns regarding heteroscedasticity. Therefore, we apply several non-linear models to the prediction of deposit outflow in the following section to address this issue.

5.2 Polynomial Regression

Moving beyond linearity, we also considered polynomial regression to assess if there is a non-linear relationship between the present within the data set. Polynomial transformation were applied to our main predictors FFR and 10y nom only. Other control variables were preserved in the model.

Then, 5-fold cross validation were used to determine the optimal order of the polynomials. After considering all possible combinations of the polynomial orders from 1 to 5, the 5-fold cross validation chose the following model:

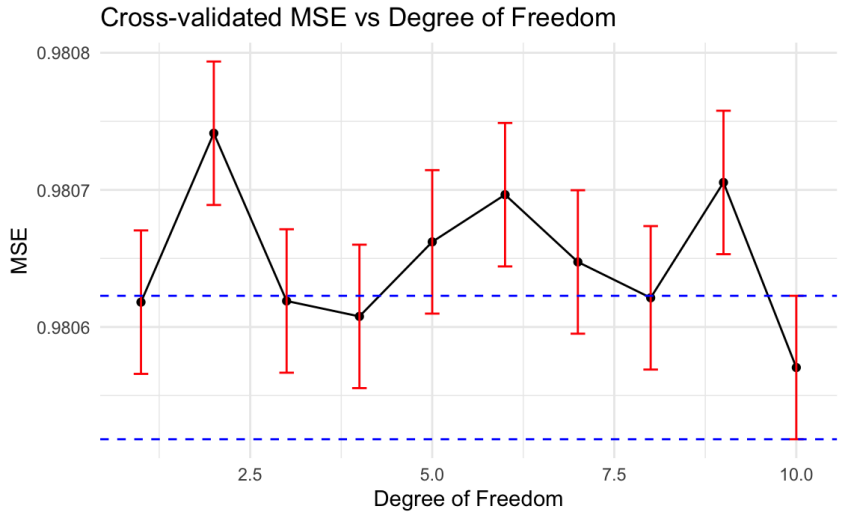
$$\log(\text{Total Deposit}_{it}) = \beta_0 + \beta_1 \text{FFR}_t + \beta_2 \text{FFR}_t^2 + \beta_3 \text{10-year int}_t + \sum_{k=1}^{p} \beta_k \text{Other Variables}_{ikt} + \epsilon_{it}, \tag{2}$$

The above model with the lowest CV error incorporates the quadratic transformation on FFR_t, and the estimated coefficient for FFR_t^2 is 3.899, which aligns with the U-shaped relationship mentioned in the literature review. However, the estimated coefficient for FFR_t^2 is not statistically significant. Therefore, we were unable to draw a definitive conclusion regarding whether our research supports the U-shaped relationship between interest rates and deposits proposed by economist Rafael Repullo. Thus, this encouraged us to continue testing other statistical machine learning techniques.

5.3 Natural Cubic Splines

Instead of fitting polynomials with higher degree, we examined the performance of natural cubic splines (for the predictors FFR_t and 10y nom) on the prediction of deposit outflows. Usually, natural cubic splines outperform polynomial regression not only because they produce more stable estimates but also because they enhance flexibility by increasing the number of knots while keeping the degree constant to avoid overfitting.

5-fold cross validation was also conducted to choose the degree of freedom for the cubic splines. The following plot depicted the cross validated mean squared errors and their corresponding degree of freedom:



The red bars on the graph are the one-standard-deviation ranges for the cross-validated MSE's. The natural cubic spline with 10 degrees of freedom has the lowest MSE. Nevertheless, in the aim of choosing the most parsimonious model, we took into account the one-standard-error rule: select the simplest model whose MSE falls within one standard deviation of the minimum MSE. The blue

dashed lines depicts the region within one standard deviation of the minimum MSE. We observed that 1, 3, 4, and 8 degrees of freedom fall into that region.

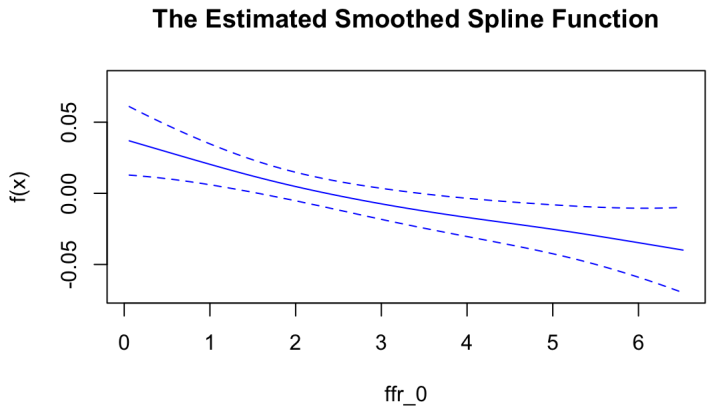
Instead of strictly adhering to the one-standard-error rule by selecting df = 1, we selected df = 4 as it has the lowest MSE among them while still being sufficiently parsimonious.

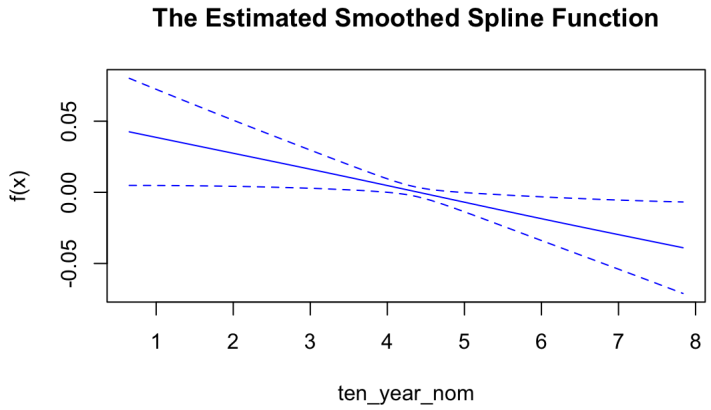
5.4 Generalized Additive Models (GAMs)

GAM is a natural way to extend the multiple linear regression model. In order to better interpret the non-linear relationships between the interest rates and the total deposits, we replaced FFR and 10y nom in Equation (1) with smoothing spline functions, while other control variables in Equation (1) remain unchanged.

In R, the gam() function in mgcv package allows us to specify the smoothing parameter estimation. In this study, we applied Restricted Maximum Likelihood Estimation (REML) to the smoothing parameter.

Two plots below display the smoothing spline functions for FFR and 10y nom respectively:

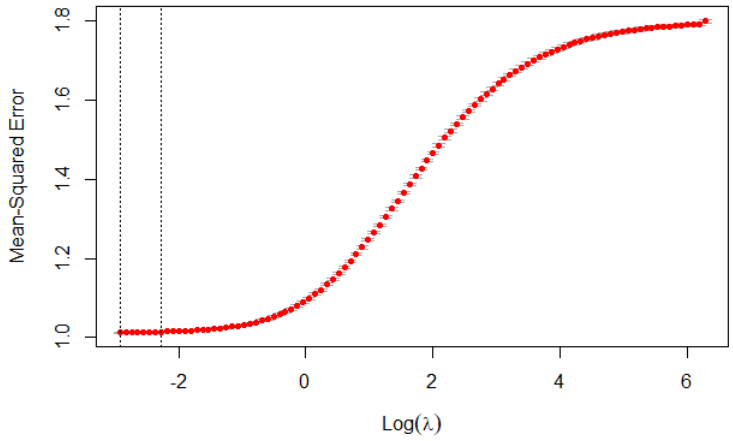




Consistent with previous models, the negative relationship between interest rates and deposit levels is also captured by the smoothing spline in the GAM.

5.5 Ridge Regression

To reduce the variance inflated by multicollinearity, we also performed ridge regression on the training dataset to predict the deposit outflow. Similar to previous models, we conducted 10-fold cross-validation for ridge regression to select the optimal value for the penalty term λ. We plotted the cross-validated MSE against the $\log(\lambda)$.



The lambda that minimizes the cross-validated MSE is: $\lambda = 0.05366196$.

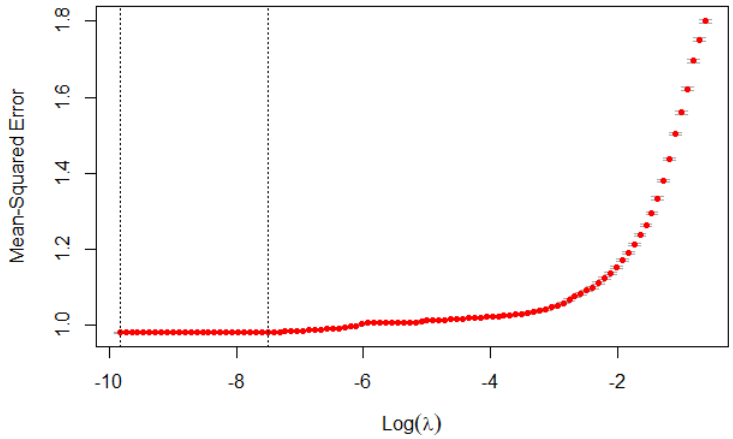
However, we chose the best lambda inside the one-standard-error interval, $\lambda = 0.10291882$, for the ridge regression model.

We compared coefficients estimated by the ridge regression model with the best lambda and with

the coefficients estimated by the linear regression model. In the ridge regression model, 41 of 57 estimates are smaller in magnitude. For example, according to the ridge regression, all else held constant, a one-percentage-point increase in federal funds rate and 10-year interest rate decrease total deposits by 0.993% and 0.935% respectively, which are less than the 1.10% decrease proposed by the linear regression model.

5.6 The Lasso

For the Lasso regression, we use the same methodology we applied to the ridge regression. We use 10-fold cross validation to select the optimal value of lambda. We also plotted the cross-validated MSE against the $\log(\lambda)$.



The lambda that minimizes the cross-validated MSE is: $\lambda = 5.3661961e-05$.

However, we chose the best lambda inside the one-standard-error interval, $\lambda = 0.0005492468$, for the lasso model.

We also compared coefficients estimated by the lasso with the best lambda and with the coefficients estimated by the linear regression model. According to our model the effect of a one percent increase the 10-year interest rate is zero, smaller than the one estimated by the linear regression model. However, a one percent increase in FFR has a strong negative effect, causing the amount of deposits to decrease by 1.339%, larger than the 1.10% decrease proposed by the linear regression model.

5.7 Principal Component Regression

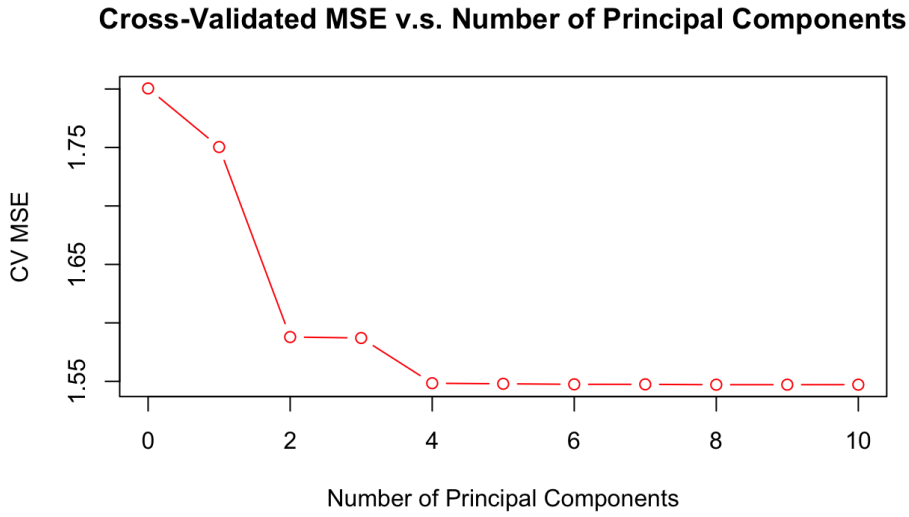
In this study, we also conducted a Principal Components Analysis (PCA) on the regression problem described in Equation (1) to investigate the relationship between the logarithm of total deposits and various economic indicators. The main goal of applying PCA was to reduce the dimensionality of the dataset and to identify the underlying structure in the data by extracting the most important

features.

It is important to note that PCA was only applied to the quantitative variables in Equation (1). PCA is preferred for a number of reasons. Firstly, it can help eliminate multicollinearity among the explanatory variables as mentioned before, which can lead to unstable parameter estimates and affect the interpretation of the results. Secondly, PCA allows for more efficient computation by reducing the number of variables, which can be particularly helpful in large datasets. Lastly, PCA can help in visualizing the structure of the data, making it easier to identify patterns and trends.

We used 5-fold cross validation to choose the optimal number of principal components to be included in the Principal Regression Model.

The following diagram displays the cross-validated MSE versus the number of principal components:



From the diagram above, the principal regression model with 8 components yields the lowest cross-validated MSE. Regressing log(Total Deposits) on the 8 principal components $Z_1, \ldots, Z_8$ can provide insights into the underlying structure of the data and simplify the regression analysis. To interpret the results in terms of the original predictors, we can recover the coefficients of the original predictors. This can be done by combining the coefficients obtained from the PCA regression with the loadings of the principal components.

Below is the table summarizing the effective coefficients of the original predictors for the PCR model with 8 components:

The table above provides insights into the relationships between the predictors and the response variable, log(Total Deposits).The coefficient for FFR is -0.0224. This indicates that, holding all other variables constant, a 1-unit increase in FFR is associated with a 2.24% decrease in the total deposits. The coefficient for 10y_nom is -0.0108. This implies that a 1-unit increase in the 10-year nominal interest rate, while keeping all other variables constant, results in a 1.08% decrease in the

Table 3: Effective Coefficients of the Original Predictors for the PCR Model with 8 Components

Predictor	**log(Total_Deposits)**
FFR	-0.0224
year	0.0228
10y_nom	-0.0108
offdom	0.4350
offfor	0.0296
Umich_consumer_sent	-0.0024
Real_disposable_income	-0.0154
Population	-0.0072
total_bank_equity_capital	0.1474
NASDAQ	0.0400

total deposits.

6 Conclusion

We compared the performance of seven different models by calculating the Test MSE for each model after refitting them on the testing dataset. The testing dataset contains half of the banks in each quarter from December 1992 to June 2022. Lower Test MSE values indicate better predictive performance. As shown in the table, Ridge Regression outperforms the other models with the lowest Test MSE of 1.9533. This suggests that Ridge Regression provides the best predictive accuracy among the considered models for the given dataset. On the other hand, Principal Component Regression exhibits the highest Test MSE of 2.0239, indicating relatively weaker performance in comparison to the other models. This comparison helps in selecting the most suitable model for predicting the Total Deposits of banks based on various factors and highlights the importance of validating model performance on an independent testing dataset.

There are several reasons why Ridge Regression may outperform other models and why Principal Component Regression (PCR) may be the weakest in predicting total deposits of banks based on various factors. In this particular problem, it is possible that all features contribute valuable information, and Ridge Regression's ability to preserve all features gives it a performance advantage over the Lasso. Besides, Ridge Regression makes fewer assumptions about the functional form of the relationship between predictors and the response variable. In comparison, Natural Cubic Spline, Polynomial Regression, and Generalized Additive Model assume specific functional forms, which may not accurately represent the true underlying relationship. Therefore, Ridge Regression can result in better predictive performance when the true relationship is not well-captured by the assumed functional forms.Besides, since Multiple Linear Regression does not include any regularization, which may lead to overfitting when there are collinear predictors or a large number of features. Ridge Regression's L2 penalty can effectively address multicollinearity issues and reduce overfitting, resulting in better generalization to the test dataset. Finally, the main limitation of PCR is that it assumes that the principal components with the largest variance explain the most variation in the response variable. However, this assumption may not always hold true. In this particular problem, it is possible that the principal components with smaller variances are more relevant to the response variable, leading to weaker predictive performance compared to other models.

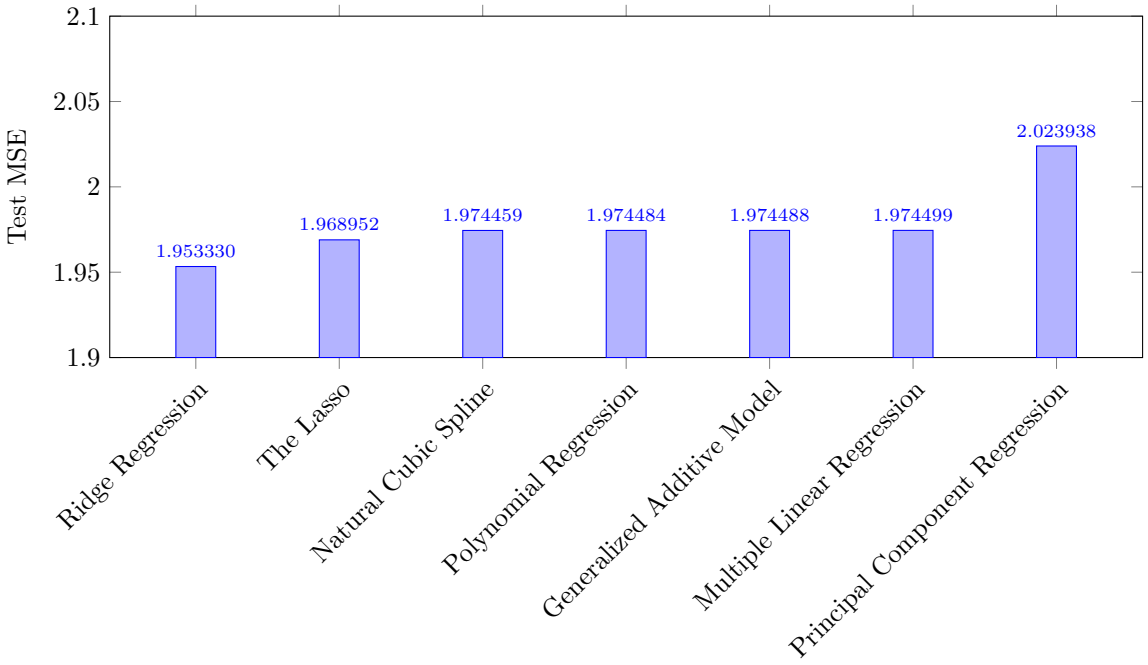


Figure 1: Test Mean Squared Error (MSE) for Different Models

In summary, we explored the impact of interest rates on bank deposit levels in the United States from December 1992 to June 2022. Employing a variety of machine learning models, we consistently found a negative relationship between interest rates and deposit levels, highlighting the fact that deposit outflows occur in high-interest-rate environments. Among the seven models tested, Ridge Regression exhibited the best predictive performance, as evidenced by the lowest Test MSE value. These findings contribute to our understanding of the dynamics of deposit outflows in response to changes in interest rates and can inform strategies to enhance the resilience of the U.S. banking sector in the face of interest rate fluctuations.

Our study is particularly timely given the current economic landscape, which is experiencing a high inflationary period that may result in even tighter credit conditions if inflation at the current rate persists. The Fed's current approach is aimed at macroeconomic and financial stabilization, but recent events, such as the failure of Silicon Valley Bank due to depositors withdrawing funds, created liquidity problems for the bank and spread fear across the banking sector. This growing lack of confidence in the economy and safety of banking institutions can lead to significant adverse effects on the broader market.

By understanding the dynamics of deposit outflows in high-interest-rate environments, banks and regulators can better prepare for and respond to deposit outflows, ultimately contributing to the stability and resilience of the U.S. banking sector in the face of changing interest rates. Our study offers valuable insights that can help inform policies and strategies to mitigate the risks associated with deposit outflows and enhance the overall health of the banking sector.

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